

# MAZEN BAHGAT

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## EXPERIENCE

### Microsoft

August 2025 - Present

*Software Engineer*

*Redmond, WA*

- Architected a multi-agent data intelligence platform on Azure AI Foundry, saving **3,000+** manual hours as LLM agents run sentiment analysis over a data pipeline of public and internal sources to flag AI data center delivery risks.
- Engineered a data platform onboarding **36,000+** site operators, turning physical site constraints into AI rack roadmaps.
- Cut GPU feasibility cycle time by **20%** by building partner-facing and internal portals where lease providers and Microsoft teams share AI cluster designs, integrated with upstream cluster design and lifecycle tracking.

### Microsoft

May 2024 - July 2024

*Software Engineering Intern*

*Redmond, WA*

- Shipped event-driven partner-engagement tracking and milestone notifications in Dynamics 365 for data center lease acquisition, cutting operational costs by **75%**; ran knowledge transfers so other teams reused its components.
- Eliminated certificate-based authentication from a production service by migrating it to Azure Managed Identity with Federated Identity Credentials, leaving zero stored credentials to rotate, leak, or expire.

### NCR Voyix

January 2024 - April 2024

*Software Developer*

*Waterloo, ON*

- Developed an NGINX/NJS proxy that decrypts envelope-encrypted (DEK/MEK) content from multi-cloud storage, securing teller-customer collaboration for **600+** branches and **1,000+** tellers; 2 other teams adopted it.
- Replaced **60-second** HTTP polling with WebSocket streaming in the React teller app, cutting data refresh latency to under 100 ms for money transfers and interbranch cash shipments.

### Microsoft

May 2023 - August 2023

*Software Engineering Intern*

*Redmond, WA*

- Migrated **6** services to OpenTelemetry with zero telemetry data loss by running old and new pipelines side by side.
- Automated intake of hyperscale data center power-capacity requirements via event-driven Azure Service Bus and Azure Functions, scoping evolving requirements with the PM and halving processing from **1,800 to 900** hours/year.

### NCR Corporation

September 2022 - April 2023

*Back-End Developer*

*Waterloo, ON*

- Implemented scalable, containerized Spring Boot and Node.js banking microservices on Kubernetes (GCP) integrated through Kafka, using feature flags and managed deployments to customize services for **6** financial institutions.
- Consolidated duplicate same-type requests into one, cutting Teller Cash Recycler client-hardware calls by **66%**.

## TECHNICAL SKILLS

- **Programming Languages:** C#, Java, Python, TypeScript, JavaScript, SQL, C, Bash, HTML/CSS.
- **Backend & Data:** Distributed systems, cloud architecture, microservices, PostgreSQL, NoSQL (MongoDB, Redis).
- **Cloud & DevOps:** Azure, GCP, Kubernetes, Docker, Bicep (IaC), CI/CD (GitHub Actions, Azure DevOps), Git.
- **Frameworks & AI:** .NET, Node.js, React, Next.js, React Native, Azure AI Foundry, LLM agents, MCP servers.
- **Practices:** SDLC, system design, design patterns, code reviews, automated testing, observability, AI-assisted dev.

## PROJECTS

**Switchback** | *TypeScript, Next.js, React Native, tRPC, PostGIS, Azure Functions, Bicep*

July 2026 - Present

- Built a full-stack trail discovery app for web (PWA) and iOS covering **47K+** trails on a shared tRPC API.
- Re-architected OpenStreetMap ingestion from a once-daily cron (4 tiles/day vs. a **14,320**-tile backlog) into a fault-tolerant Azure Service Bus and Functions pipeline with idempotent jobs, lifting throughput **125x**.
- Diagnosed a CPU-bound ingest hotspot and cut a dense tile's spatial matching from **530 s to 0.8 s** with an in-memory grid index, benchmarked against a PostGIS spatial join with byte-identical output.
- Provisioned Azure infrastructure in Bicep with keyless queue/storage access via managed identities and **11** alerts.

## EDUCATION

**University of Guelph**

September 2020 - April 2025

*Bachelor of Computing, Honors Co-op, Major in Software Engineering, Minor in Marketing*

*Guelph, ON*

- **Coursework:** Data Structures, Algorithms, Operating Systems, Object-Oriented Programming, Software Testing.